(a) Motivating application: TrackMania 2020



Real-time visual control

Goal: learn new conditions while retaining earlier control behavior.

Saved development capture

(b) Executed study: HalfCheetah-v5 and Ant-v5

16 source policies per task
2 controllers x 8 source seeds

Old condition A: clean
2 replay episodes per source

Five branches per source
Noise or 3-step delay x $\rho \in \{0, 0.5\}$
plus a clean-training control

Continued SAC updates
20,000 new steps, 5,000 updates
ρ = old-replay batch fraction

Freeze the updated checkpoint
At 0, 2k, 4k, 8k, 12k, 16k and 20k new steps

Evaluate A: retention
Clean return; critic residual

Evaluate B: adaptation
Sensor return; critic residual

Same checkpoint for A and B; evaluation episodes stay out of replay.